Benjamin Osafo Agyare

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EDUCATION

Aug 2021 - May 2026	PhD in Statistics, **University of Michigan**, Ann Arbor Thesis title: Distributional Learning via Flexible Expectile Regression: Methods for Dependent, Multivariate and Incomplete Data Advisor: Prof. Kerby Shedden
Aug 2019 - July 2021	MS in Statistics & Data Science, **University of Nevada**, Reno Recipient of Full Scholarship from the Department of Mathematics and Statistics in $ 75,000+ (USD).
Sept 2013 - May 2017	BS in Actuarial Science, **Kwame Nkrumah University of Science and Tech** Honors Thesis: A Survival Analysis on the Surrender of Life Insurance Policies in Ghana. Advisor: Prof. Gabriel Asare Okyere

SKILLS

Programming/Tools:	Python, R, SQL, PyTorch, TensorFlow, Scikit-Learn, PySpark, HTML, PHP, Javascript, git, Slurm, Terminal, Markdown
AI/ML:	Deep Learning (NLP, CV), XGBoost, SVM, Logistic, Trees, Bayesian methods, A/B Testing, Causal Inference
Statistics:	Regression, Survival, Mixed Models, GEE, SEM, High-Dimensional (Ridge, LASSO, ENet, PCA), Experimentation

WORK EXPERIENCE

June 2025 - Aug 2025 — **AI & Data Science Summer Associate**, JPMorganChase, Wilmington, DE

- Developed and benchmarked advanced multi-class credit risk models—including XGBoost, neural networks, and RKHS-based kernel methods—to uplift card acquisition performance beyond legacy binary models.
- Engineered scalable training pipelines for 19M+ customers and 500+ features, optimizing for GPU and parallel CPU infrastructure while ensuring model explainability with SHAP for regulatory compliance (reason/adverse action codes).
- Delivered production-grade models with AUC $\approx$ 96% and KS $\approx$ 80%, evaluated via lift charts across in-sample, holdout, and out-of-time datasets to support downstream decisioning by the credit strategy team.

May 2024 - Aug 2024 — **Research Intern**, Center for Global Health Equity, University of Michigan, Ann Arbor, MI

- Developed diagnostic network optimization models to enhance laboratory networks in low- and middle-income countries, aiming to reduce disease burdens and mitigate future pandemics.
- Built and calibrated generalizable agent-based simulation models for diagnosing and managing epidemic-prone diseases and other public health concerns in developing countries.

Aug 2021 - Present — **Graduate Researcher**, University of Michigan, Ann Arbor, MI

- Develop flexible, distribution-free, and robust statistical methods for characterizing the full conditional distribution of heterogeneous data and covariate-dependent errors.
- Implement efficient computational algorithms to enhance the accuracy and reliability of statistical inferences
- Presented research findings at top conferences and symposiums, earning awards for outstanding research in statistical methodologies.

May 2020 - Aug 2020 — **Predictive Modeling Intern**, Employers Insurance Group, Reno, NV

- Conducted Territorial Analysis on claim frequencies by employing Spatially Constrained Clustering Algorithms and Generalized Additive Models, leading to the re-clustering of rating territories for enhanced pricing models.
- Developed Loss Development Models utilizing Elastic-Net Poisson GLM, significantly enhancing the predictive accuracy of future losses and optimizing reserve setting for the company.
- Constructed Pure Premium models through GLMs and Zero-Inflated Models for accurate prediction of future loss costs.

SELECTED RESEARCH AND PROJECTS

- **A Distributed Optimization Package for R**
  - Authored an R package for distributed optimization, implementing algorithms where a global objective function, expressed as a sum of local objective functions assigned to agents (e.g., nodes in a computer network), is optimized through collaboration, with experimentation showcasing its efficacy in solving distributed and big data statistical problems (GitHub link).
- **Enhancing Computational Efficiency in High-Dimensional Bayesian Analysis: Applications to Cancer Genomics**
  - Advanced the application of Bayesian shrinkage techniques by rigorously benchmarking Two-Block vs. Three-Block Gibbs samplers, demonstrating faster convergence and scalability in high-dimensional settings; applied to NCI-60 cancer genomics data to predict protein expression and identify key genes driving cancer pathways (link).
- **Hierarchical Classification for Predicting Metastasis Using Elastic-Net Regularization on Gene Expression Data**
  - Designed and implemented a novel elastic net–based hierarchical classifier, applied to gene expression data, achieving 97% tissue-of-origin and 90% metastasis prediction accuracy, while uncovering a negative link between mitochondrial gene expression and metastasis (link).

SELECTED HONORS AND AWARDS

May 2025	**Rackham Doctoral Student Intern Fellowship**, The University of Michigan.
May 2024	**Best Poster Award**, The Michigan Student Symposium for Interdisciplinary Statistical Sciences (MSSISS).
May 2024	**Outstanding Graduate Instructor Award**, - *Honorable Mention*, Dept. of Statistics, The University of Michigan.
April 2022	**1st Place**, Capstone Project Competition in Statistical Learning, The University of Michigan.

TEACHING AND MENTORSHIP

Research Mentor: Co-supervise four (4) undergraduate students on projects including *Robust Matrix Factorization techniques for analyzing biomedical data* (ArXiv link), and *Vector Embeddings for irregular longitudinal data via U-Statistics and Bernstein Polynomials*.

Graduate Instructor: Develop instructional content, conduct labs, and evaluate assessments for 1,000+ students across 10+ graduate and undergraduate classes, including in Computational Statistics, Regression, GLMs & Mixed Models, and Semi-Parametric Models.

SELECTED LEADERSHIP AND SERVICE

Member	Computing Committee, Dept. of Stats, University of Michigan	Sept 2023 - Present
Member	Recruitment & Admissions Committee, Dept. of Stats, University of Michigan	Jan 2022 - Present